



Soft sensors based on interpretable learners for industrial-scale fed-batch fermentation: Learning from simulations

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ARTICLE INFO

Keywords:

Interpretable learner
Soft sensor
Penicillin production
Process monitoring
IndPenSim model

ABSTRACT

Monitoring bioprocesses is a challenging task where most of the variables of interest can only be measured offline. Soft sensors have emerged as a solution to provide online estimations. This work compares interpretable learners such as CART, M5, CUBIST, and Random Forest as soft sensors for industrial-scale fed-batch fermentation of penicillin production. A structured model of industrial-scale penicillin fermentation is implemented to generate the dataset and train the interpretable learners. Variables such as substrate feed rate, agitation, temperature, pH, dissolved oxygen, vessel volume, CO₂, and O₂ percent in off-gas are considered as independent (predictors). The CUBIST model has achieved the best results with values of 0.908, 9.916, 3.149, and 1.920 for the coefficient of determination, Mean Squared Error, Root Mean Squared Error, and Mean Absolute Error, respectively. These results demonstrate the feasibility of developing soft sensors based on interpretable models to predict penicillin concentration at an industrial scale.

1. Introduction

The transition toward Industry 4.0 calls for the development of smart factories that consider process automation and their adaptability to handle different scenarios (Yang and Gu, 2021). Autonomous operation requires robust monitoring and control strategies that operate in real time (Narayanan et al., 2020; Rathore et al., 2022). Monitoring of bioprocesses is a challenging task because most of the critical quality attributes (CQAs) can only be measured off-line (Kadlec et al., 2009; Rathore et al., 2022). Soft sensors have emerged as a solution for providing on-line estimates of process variables which are measured either with low sampling rates or by off-line analysis (Ji et al., 2023). A soft sensor, also known as a virtual or software sensor, is a computational model or algorithm that estimates or predicts a physical process or variable in a system using data of available measurements. Unlike traditional sensors, which are physical devices that directly measure variables or parameters, soft sensors use algorithms to infer or predict the values of parameters that are difficult or expensive to measure directly (Fortuna et al., 2007; Kadlec et al., 2009). These algorithms can be model-based or data-driven. Model-based approaches are based on

first principles of physical or chemical laws that could provide detailed process descriptions. However, they require robust models. Data-driven approaches rely on large datasets and the use of machine learning techniques to learn patterns and relationships from historical data that will allow them to adapt and improve as they receive more data (Yan et al., 2017). In this type of approach, process models are not required and the process can be treated as a black box (Rathore et al., 2022).

Biopharmaceutical industry is one of the most advanced bio-industries (Erickson et al., 2021). It faces, however, the need to increase market demand, fulfill stringent quality regulations as well as to digitalize and automate its procedures (Mitra and Murthy, 2022).

The integration of soft sensors in the biopharmaceutical industry contributes to improve processes, increased efficiency, and enhance product quality (Narayanan et al., 2019b). As the industry continues to advance, the application of soft sensors becomes increasingly valuable for achieving consistent and high-quality biopharmaceutical production.

In the biopharmaceutical industry, soft sensors play a crucial role in process monitoring and control, ensuring the quality and efficiency of production. For instance, soft sensors are used to monitor key param-

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ters in bioreactors, such as cell density, viability, and product concentration. Leveraging mathematical models and data-driven approaches, these sensors help to optimize the growth conditions of cells and the production of biopharmaceuticals. For example, Wang et al. (2020) proposed a multi-output least square support vector (MLSSVM) soft-sensor to estimate key biochemical variables of the fermentation such as the concentrations of biomass, substrate, and l-lysine using temperature, pH, speed of stirring motor, dissolved oxygen, acceleration rate of glucose flow, acceleration rate of ammonia flow, and air flow rate as input variables. They have used the data of ten batches to test the modeling with the MLSSVM. Then, Tokuyama et al. (2021) developed soft sensor using 24 machine learning algorithms. The authors used all available variables (27 in total) corresponding to measured variables such as CO_2 or O_2 concentration in exhaust gas, pH, or culture time; control variables such as feeding rate of sugar or ammonium, regulation valve for feeding sugar source or total feed volume, and theoretical values like theoretical sugar consumption rate. Their work highlighted that soft sensor based on nonlinear models using ensemble learning showed higher prediction accuracy. Despite their results, the authors stressed that there is one common issue that affects model performance: noise in the input data. However, simple models such as decision trees and random forests still yielded interesting results. In this latest study, several issues were identified such as computational cost, especially with nonlinear models, and the amount of available data.

Soft sensors are also employed to estimate pH and temperature in bioreactors. Maintaining precise pH and temperature levels is crucial for the stability and activity of biological processes. Furthermore, soft sensors help predict the performance of biopharmaceutical products (Narayanan et al., 2019a). The algorithms behind the soft sensors could also be employed to forecast the amount of product that can be obtained based on historical data, which allows for better planning and resource allocation. Additionally, the models used by these sensors can be adapted to different scales, ensuring consistency and efficiency as production volumes change.

A key biopharmaceutical product that has been studied in this domain is penicillin. The application of soft sensors in penicillin production, which involves complex biochemical processes, allows for real-time monitoring, control, and optimization of the fermentation process, ultimately leading to improved yields, reduced production costs, and consistent product quality. These sensors play a critical role in ensuring the efficiency and reliability of the penicillin manufacturing process. These sensors help maintain optimal conditions for the growth of the penicillin-producing microorganisms, typically strains of the fungus *Penicillium chrysogenum*.

Moreover, soft sensors are often integrated with on-line analytical techniques, such as near-infrared spectroscopy or mass spectrometry (Brunner et al., 2021). These techniques provide real-time information on the concentration of penicillin and other relevant components in the fermentation broth.

The construction of soft sensors for this application has typically been based on data obtained from a model for penicillin production by the fungus *Penicillium chrysogenum* developed by Goldrick et al. (2015). This model predicts all on-line and off-line process measurements including off-gas analyses. The model has been converted into a simulation tool called *IndPenSim* from which 100 batches have been simulated and reproduced (Goldrick et al., 2019). As *IndPenSim* can provide simulated data, it has become a prevalent resource for studying industrial-scale penicillin fermentation processes via machine learning. The data set generated from *IndPenSim* includes 100 batches. The first 90 batches were obtained under normal operating conditions using three different control strategies, and the last 10 batches contained faults, resulting in process deviations (Goldrick et al., 2019). For instance, *IndPenSim* data sets have been used to build a multilayer perceptron model to infer the concentration of phenylacetic acid (PAA) using batches 80 and 81 for model training and testing, respectively (Gao and Shardt, 2022). These batches were controlled with an advanced process

control strategy using Raman spectroscopy.

Using the same data set, Alexandre et al. (2023) proposed a methodology based on tri-clustering algorithms to assess the statistical significance of patterns in tensor data, which are multidimensional arrays. Tensor data structures enable the exploration for uncovering novel patterns and relationships that cannot be obtained from tabular data alone. Gu et al. (2023) proposed a subspace identification combined with a common feature learning scheme for modeling *IndPenSim* batch processes. This approach used different batches to train the state-space model and extract common feature parameter matrices. Historical batch data were projected into the common feature space to reduce sample requirements for training new models.

In the context of fault detection and diagnosis, *IndPenSim* has been used as a benchmark to perform investigations with autoencoder neural networks (AE-NN). For instance, Multiway Partial Least Squares Autoencoder (MPLS-AE) was proposed as a process monitoring approach (Agarwal et al., 2022). This autoencoder was trained with a novel objective function, tailored for fault detections. The main objective was to learn the distribution of batches without faults in order to differentiate from the batches with process deviations. Hematillake et al. (2022) have applied a supervised deep convolutional autoencoder for fault detection and diagnosis in a penicillin fed-batch process. This study proposed an objective function based on the effect of bioreactor design parameters on both productivity and loss of process profitability. Although accurate, neural networks are complex and lack interpretability making it difficult to identify relationships between features. Additionally, experimental data obtained from biological processes may contain noise and outliers that can have a negative effect on the training of such models (Pang et al., 2022). Similarly, Aghaee et al. (2023) have very recently proposed an unsupervised timewise long short-term memory autoencoder (LSTM-AE) network to identify and diagnose faults in batch processes. This method has shown the regions with normal operation in batches with faults. However, the complexity of the model hinders the ability to find dynamic correlations and select objective functions for training such models (Aghaee et al., 2023).

So far, recent machine learning techniques have been applied to predict penicillin concentration from the *IndPenSim* data set. For instance, Kjærsgaard et al. (2021) proposed a sampling approach to adjust imbalanced data with the goal of improving the performance of a linear regression model trained on penicillin production. The goal was to down sample the covariates in the training data and subsequently fit a linear regression. The authors investigated the use of three sampling approaches to adjust for imbalance. All of the approaches were based on principal component analysis (PCA) and partial least squares (PLS) techniques. Nevertheless, conventional PCA and PLS are linear procedures and therefore of limited effectiveness when applied to nonlinear batch fermentation problems (Lennox et al., 2001). Ji et al. (2023) have very recently proposed a deep learning model considering a moving window with a fixed-length time steps, and if the window is at the end of the batch, it will move from the beginning of the next batch to preserve the internal information of original batch data. Thirty batches with normal operating conditions were used for model training, and the other 60 batches were used as testing data. Even if deep learning algorithms are known for their prediction accuracy, they come with the trade-off of transparency and interpretability. Deep learning models tend to be black boxes with functions that are too complicated for humans to understand (Rudin, 2019).

In this context, interpretable learners such as decision trees, additive models, and sparse linear models, create models that can be directly inspected to understand model predictions (e.g., the weight of a feature in a linear model, a path in a decision tree, or a specific rule). These learners employ a relatively limited set of internal elements (i.e., paths, rules, or features), but they offer traceability and transparency into the decision-making process. As long as these approaches maintain prediction accuracy, they provide insight into the decisions made by the model (Rai, 2020).

This work proposes to study the performance of interpretable learners using the *IndPenSim* data set. The goal is to compare interpretable learners, such as CART, M5, CUBIST decision trees, and Random Forest, in terms of prediction accuracy and interpretability to provide insights about the use of soft sensors for the penicillin production industry. Therefore, this paper addresses the three following objectives:

1. To provide an explanation of data through rules by relating the behavior of process variables with the types of regressions obtained by the machine learning algorithms.
2. To verify the performance of four algorithms considered as interpretable learners: CART, M5, CUBIST, and Random Forest in the prediction of penicillin production by using the dataset of *IndPenSim*.
3. To create a parsimonious soft sensor based solely on-line measurement to explain penicillin production from the *IndPenSim* dataset.

The remainder of this paper is organized as follows. Section 2 briefly describes the *IndPenSim* dataset and outlines the theoretical basis of the interpretable learners. Section 3 is dedicated to explain the construction of the interpretable models, offering a comprehensive breakdown of the components as weight of features, decision tree paths and rules. The results and discussion are presented in Section 4. Finally, conclusions and future work are presented in Section 5.

2. Material and methods

This section describes the *IndPenSim* data set used to build the interpretable learners. It details each batch, the involved process variables, their respective ranges, and the distribution of the dataset for training and testing. Finally, it describes three randomly selected instances from batch 2, allowing to study the capability of the trained models.

2.1. *IndPenSim* dataset

The data used in this work has been simulated using the structured model for industrial-scale penicillin fermentation (*IndPenSim*) to represent penicillin production from mixed substrates (Goldrick et al., 2019). A minimum production of 2000 kg of penicillin per batch was fixed as a set point to consider the process efficiently. The dataset is composed of 100 batches, divided into four types according to different control strategies: (I) controlled by a recipe-driven approach, (II) controlled by operators, (III) controlled by an Advanced Process Control (APC) solution using the Raman spectroscopy, and (IV) uncontrolled containing faults resulting in process deviations:

- Control by recipe-driven approach (batches 1–30):** this control allowed to manipulate variables such as sugar feed rate (F_s) and phenylacetic acid flowrate (PAA) using a set of rules by following a driven approach in which a fixed profile was used throughout the batch.
- Control by operators (batches 31–60):** an operator manipulated the fixed profile throughout the batch (operator dependent), i.e., this control mode replicates the manual actions performed by the operators, adjusting F_s and PAA throughout the batch.
- Control by an Advanced Process Control (APC) solution using Raman spectroscopy (batches 61–90):** this control used an empirical mathematical model to simulate a realistic PAT analyzer (Raman spectroscopy device).
- Batches with faults resulting in process deviations (batches 91–100):** the *IndPenSim* model was used to simulate faults in aeration, pH sensor drifts, variations of substrate concentration, and coolant by implementing two standardized multivariate data analysis (MVDA) as fault detection algorithms.

Each simulation was developed in a 100 m^3 reactor operating at 288 K, 1 atm, and 100 rpm for an average of 230 h. Variables such as temperature (T) and pH (pH) were regulated in all strategies using a Proportional Integral Derivative (PID) controller.

The data set of *IndPenSim* contains 113,935 data points (here called instances) corresponding to different times of the processes in the 100 batches. At each instance, 2238 variables are available, where 39 correspond to process variables, and the remaining 2199 are Raman spectroscopy measurements. In this work, only process variables are considered, one being the penicillin concentration. A preselection of variables was made from the remaining 38 variables to predict penicillin concentration. In this case, variables that are measured off-line have not been considered. Fig. 1 shows the variables used to build the interpretable learners. As it can be observed, online measured independent variables are substrate feed rate (F_s), agitation (rpm), temperature (T), pH, dissolved oxygen (DO), vessel volume (V), CO_2 percent in the off-gas ($CO_{2,og}$), and O_2 percent in the off-gas ($O_{2,og}$), while the dependent variable (variable to be predicted) corresponds to Penicillin concentration (P).

The data set containing 100 batches has been used either for training or testing. The train-test split ratio was chosen as 80:20 batches for each control strategy. Table 1 outlines the distribution of training and test sets.

Table 2 shows the ranges of the variables used to build the soft sensors along the four control strategies. These ranges have been taken from the data set *IndPenSim*. It can be observed that T values are almost constant with a small variation of 2 or 3°. pH oscillates between 5.4 and 6.76. DO varies considerably from 1.0 mg/L in the growth phase to 16.5 mg/L in the production phase, which agrees with F_s that increased from around 8 to 150 L/h along the cultures. Lower values of F_s were reported for the batches with faults since it was one of the disturbed variables.

One batch (batch 2) was chosen from the test sets to illustrate and compare the results for Penicillin (P) production with the various soft sensors on the *IndPenSim* model predictions. The dynamic profiles of all the variables used in the data set are displayed in Fig. 2. In order to assess the capability of the trained models 3 instances were randomly selected throughout batch 2 at 68.4, 120, and 200 h, which corresponded to penicillin concentrations of 8.31, 19.63, and 29.10 g/L, respectively. The values of the predictors (input variables) were:

- **Instance 1 :** $T = 297.98$ K, $pH = 6.4887$, $DO = 14.786$ mg/L, agitation = 100 rpm, $O_{2,og} = 0.18824$, $CO_{2,og} = 1.6131$, $V = 76$, 213 L, $F_s = 84$ L/h
- **Instance 2 :** $T = 297.97$ K, $pH = 6.5006$, $DO = 12.3110$ mg/L, agitation = 100 rpm, $O_{2,og} = 0.18722$, $CO_{2,og} = 1.7897$, $V = 83$, 934 L, $F_s = 90$ L/h
- **Instance 3 :** $T = 297.9500$ K, $pH = 6.4883$, $DO = 12.1020$ mg/L, agitation = 100 rpm, $O_{2,og} = 0.19376$, $CO_{2,og} = 1.3944$, $V = 76$, 413 L, $F_s = 80$ L/h

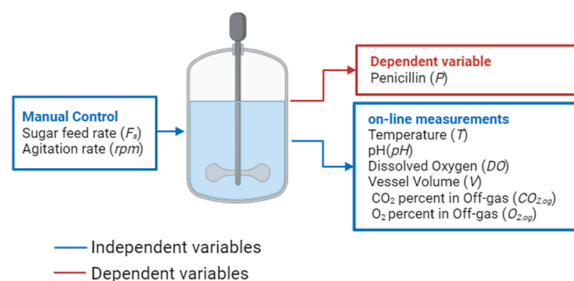


Fig. 1. Summary of the *IndPenSim* dataset variables. Image adapted from (Goldrick et al., 2019).

Table 1

Distribution of training and test set. The dataset was divided by control strategies: (i) Controlled by recipe-driven approach, (ii) Controlled by operators, (iii) Controlled by an APC solution using the Raman spectroscopy, and (iv) faults resulting in process deviations.

Control strategy	Batches used in training set	Batches used in test set	Training instances	Test instances
Controlled by recipe driven approach	21, 13, 26, 9, 29, 30, 11, 24, 3, 20, 7, 22, 17, 5, 6, 12, 10, 19, 25, 27, 4, 1, 8	2, 14, 15, 16, 18, 23, 28	26,350	7975
Controlled by operators	54, 47, 37, 45, 55, 59, 38, 50, 34, 57, 43, 44, 49, 39, 53, 33, 40, 36, 58, 51, 56, 42, 31, 48	32, 35, 41, 46, 52, 60	27,590	6795
Controlled by an APC solution using the Raman spectroscopy	65, 70, 67, 73, 82, 87, 77, 76, 90, 66, 85, 69, 89, 83, 71, 88, 81, 72, 62, 80, 61, 74, 84, 75	64, 68, 78, 79, 86, 63	26,775	6925
Faults resulting in process deviations	92, 96, 97, 94, 98, 93, 95, 99	91, 100	9085	2440

Table 2

Ranges of the variables used to build the soft sensors in the different control strategies.

Control strategy	T (K)	pH (–)	DO (mg/L)	$CO_{2,og}$ (%)	$O_{2,og}$ (%)	V (L)	F_s (L/h)	P (g/L)
Controlled by recipe driven approach	297.4–300.2	5.696–6.710	2.581–16.09	0.070–2.187	0.148–0.223	56,840–89,539	8.00–150	0–36.183
Controlled by operators	297.4–300.2	5.823–6.759	1.00–16.45	0.070–2.233	0.1584–0.2203	57,001–90,001	8.00–150	0–35.088
Controlled by an APC solution using the Raman spectroscopy	297.5–300.9	5.396–6.689	2.693–16.27	0.079–2.192	0.155–0.227	56,942–95,716	8.00–150	0–33.209
Faults resulting in process deviations	296.8–302.2	5.482–6.766	2.412–16.50	0.075–7.121	0.1295–0.2159	56,549–86,987	2–150	0.0009–32.809

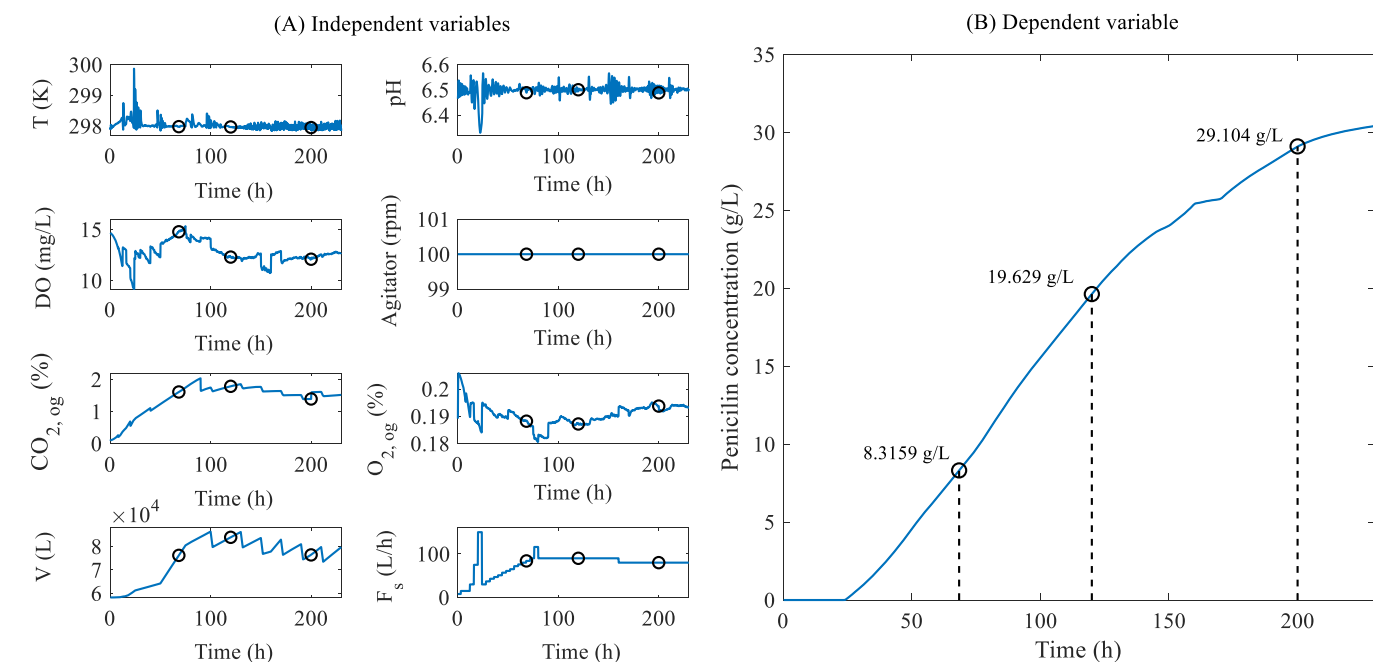


Fig. 2. Dynamic behavior of independent and dependent variables in Batch 2 estimated with the *IndPenSim* model. Back circles: instances selected throughout the batch 2.

2.2. Interpretable learners

The properties of interpretable learners broadly fall into two categories. The first category refers to transparency that provides insights into how the model works. The second category involves post-hoc explanations, particularly relevant for black box learners, which offer additional information about how a model works (Lipton, 2018). In this paper, we study four interpretable learners classified into the transparency category: CART, Random Forest, CUBIST, and M5. These learners have been selected for their ability to provide transparency through their principles of tree-based structure, rule-based representation, feature importance analysis, and model simplicity. The next section describes the four learners used for constructing interpretable learners, by utilizing the data set formulated with *IndPenSim*.

2.2.1. CART decision tree

Decision trees can be created using several learners for classification tasks like ID3 (Quinlan, 1986), C4.5 (Salzberg, 1994), and CART, which can be used for both classification and regression problems (Breiman, 2001). CART uses a greedy approach to splits the input space into several non-overlapping rectangular or cuboid regions (nodes); in a regression analysis, a constant value is assigned to each region (Ghiasi and Mohammadi, 2017). An evaluation of all the possible splits is performed where the one that best reduces the error is selected as node.

2.2.2. M5 decision tree

The M5 learner is an extension of the traditional decision tree

algorithm (Quinlan, 1992) introduced for regression tasks with linear regression functions at the terminal nodes (leaf) for the different split data portions from the root node to the leaf nodes. The tree generation model follows two steps.

1. Splitting criteria. The idea is to select the attribute having the best error reduction. The error at each node is measured based on the standard deviation (SD) reduction (SDR) (Heddam, 2021):

$$SDR = SD(T) - \sum \frac{|T_i|}{|T|} SD(T_i)$$

where T presents the number of examples reaching a node, T_i denote the subset of cases that have the i^{th} outcome of the potential test. The standard deviation of the observed values is denoted by SD . The splitting criteria is repeated recursively, often producing a large tree-like structure that may cause overfitting. Consequently, the tree must be pruned back.

2. Tree pruning. In this step, the tree is pruned replacing the subtrees with linear regression functions. This technique of generating the model tree splits the parameter space into subspaces and builds a linear regression model in each of them.

2.2.3. CUBIST

The Cubist learner is an advanced version of M5 that explores non-linear relationships in observed data. An M5 tree is created, and the paths along the decision tree are flattened into rules, which are simplified and pruned. Cubist creates intermediate linear models at each step of the M5 tree. The predictions of the linear regressions at the terminal node are smoothed by taking into account the prediction from the linear model in the previous node of the tree; this process occurs recursively up into the tree (Quinlan, 1993; Wang and Witten, 1996).

2.2.4. Random forest

Random forest (RF) builds multiple decision trees using different subsets of available features of the training data with bootstrapping and then combines their predictions to make a final prediction. Bootstrapping ensures that each decision tree in the random forest is unique in a way to reduce the overall variance of the RF model (Breiman, 2001). For the final prediction of a new instance, RF aggregates the predictions of individual trees. In regression tasks, the final prediction is typically the average of the individual tree predictions.

3. Results and discussions

The following sections show the results of the different soft sensors. Initially, a study of the various hyperparameters of the interpretable learners used to build the soft sensors is conducted. Subsequently, a comparison of the different soft sensors used for penicillin estimation is carried out. Finally, the soft sensors were tested using batches 15, 60, 86, and 91.

3.1. Hyperparameters tuning and explanation of interpretable models

This subsection presents the hyperparameters of the interpretable learners used to build the soft sensors. In addition, the rules and the decision trees that compose the models are explained. Soft sensors were built using a 10-fold cross-validation process with three repetitions. Mean Absolute Error (MAE), Squared Mean Error (SME), Root Squared Mean Error (RSME), and the coefficient of determination (R^2) were used to assess the performance of the models.

3.1.1. CART

The hyperparameter "maximum tree depth" was tuned to build the CART model. This hyperparameter refers to the maximum number of decision tree levels (rules). Setting a high value can lead to overfitting, where the model becomes complex and the decision tree works well on

the training data but poorly on new or unseen data. On the other hand, setting a low value of maximum tree depth can lead to underfitting, where the decision tree is too simple and fails to capture relevant patterns in the training set. A range of values between 1 and 10 was tested to choose the maximum tree depth. The performance was with the R^2 , as shown in Fig. A.5 in Appendix. A maximum tree depth of 7 was chosen as a good trade-off between the number of parameters and R^2 .

The construction of the CART was carried out using 4 variables that are easily measured at the industrial level: V , DO , $CO_{2,og}$, and F_s . According to Fig. 3, the algorithm sets V and DO as the most critical variables. Followed by $O_{2,og}$ and F_s . The value of P is straightforward from the given set of rules. However, the predicted values of P are limited to 8 values, with a maximum value of 26.5 g/L. This means that the prediction will not be able to achieve the largest possible values, corresponding approximately to 36 g/L (see Table 2). Regarding the rules, the first one classifies the data into two parts according to the volume: one with null to low penicillin titer and one with large titers. The first corresponds to the growth phase and the establishment of the limitation toward penicillin production, while the second one represents the exponential production of penicillin.

This interpretable model is easy to read and to apply for predictions. For example, let us consider instance 1, previously described in Section 2.1, to assess the capability of the trained model. For this instance, penicillin concentration is calculated as follows:

If $V > 68.6 \times 10^3 L$, the instance is sorted within 72 % at the root node (first node), then into the 48.9 % at the second node ($DO \geq 13.4 \text{ mg/L}$) into the third node F_s is higher than 82 L/h, so the instance is considered within 21.5 % of the data set, and finally into the 16.6 % at the fourth node. This could be translated into a mean value of penicillin concentration corresponding to 26.5 /L, which is far from the real result of 8.3159 g/L (Fig. 4A). The same procedure is used for instances 2 and 3, obtaining values of 10.7 and 16.5 g/L, respectively.

3.1.2. M5 decision tree

The recursive partitions of the M5 learner algorithm are based on feature values and create internal nodes. At a certain point, the learner stops the splitting and creates leaves. The hyperparameter "minimum number of instances per leaf" allows to setting a threshold on the number of instances required for each leaf. The value of this hyperparameter controls the tree complexity and avoids overfitting. Similar to CART learner, a smaller value for the "minimum number of instances per leaf" hyperparameter can lead to overfitting, whereas a larger value for this hyperparameter can result in a simpler decision tree with fewer levels or splits. A sequence of values between 1000 and 15,000 for the minimum number of instances per leaf was tested. A minimum number of instances per leaf of 12,000 was selected, considering that R^2 values do not present changes until 15,000 instances. Fig. A.6 in the Appendix shows the tested number of instances per leaf.

Fig. 5 depicts the M5 decision tree. During the prediction phase, a new instance crosses down the decision tree until it reaches a leaf. The M5 learner applies the regression model associated between the root node and the leaf to estimate the prediction for that instance. It is observed that the most important variable was the volume.

Each Linear Model (LM) corresponds to a linear regression model described by an independent term and coefficients for four variables $CO_{2,og}$, $O_{2,og}$, F_s and DO (Table A.1 in Appendix). It is worth noting that V is a good separator for the LM, but it does not take part in the linear regressions. From those models, it could be observed that F_s is only important when large independent terms are reported. Moreover, the rest of the variables adjust the predicted value within less than ± 1 g/L of the independent term. Therefore, the maximum titer of penicillin that could be predicted with this interpretable learner is around 28.486 ± 0.5 /L, which will follow LM 5.

For instance, if it is considered a volume lower than 63,411 L, penicillin production is low and according to $CO_{2,og}$ either regression

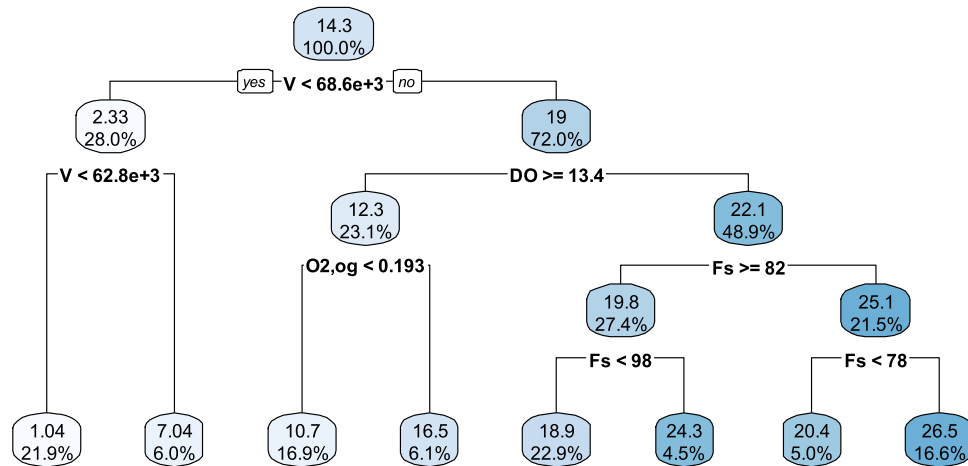


Fig. 3. CART decision tree model (Max tree depth = 7).

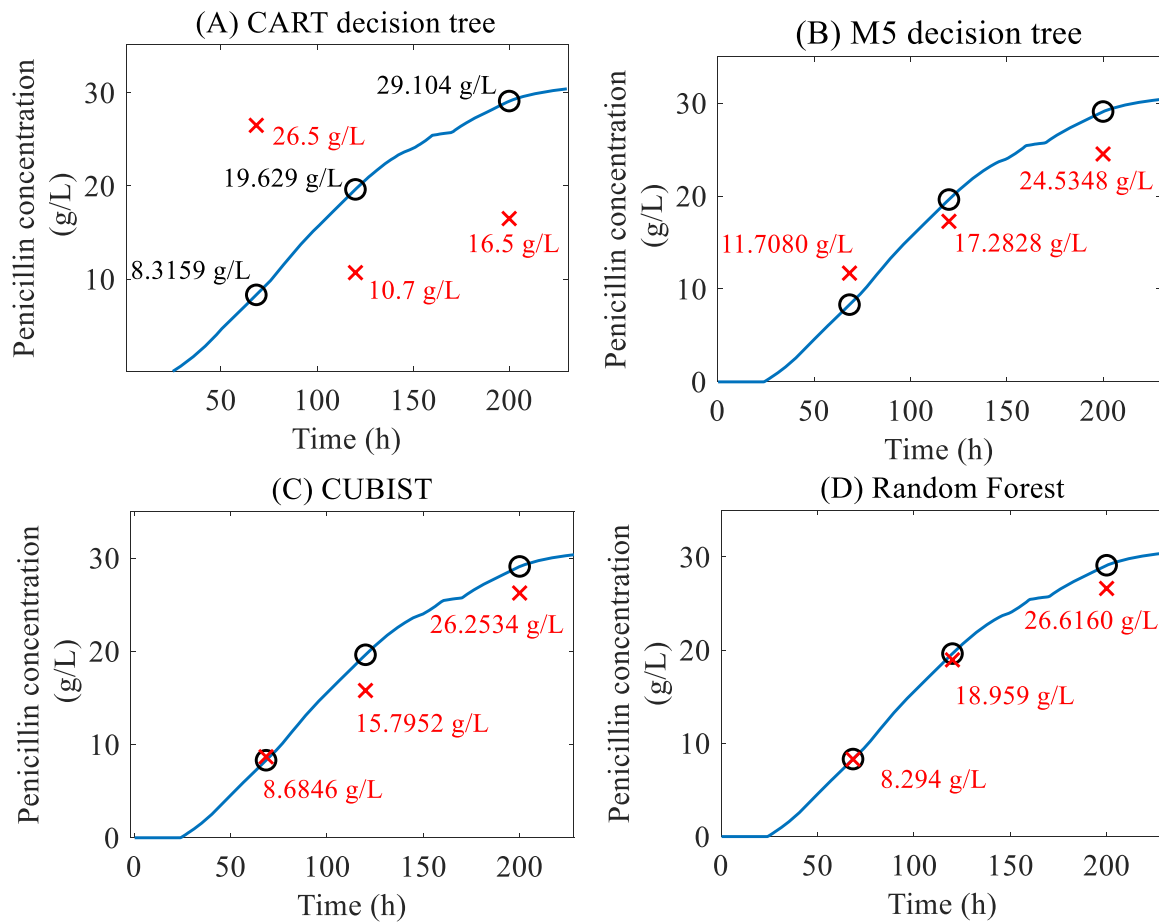


Fig. 4. Soft sensors estimations. Back circles: instances selected throughout the batch 2, red x: soft sensors predictions.

LM 1 or LM 2 is chosen for the prediction. In these two regressions, however, the only difference is the independent term, leading to estimate a larger value of penicillin in LM 2 than in LM 1. This could be interpreted as more CO_2 at the beginning of the culture higher penicillin concentration. This can be confirmed by Fig. 2 (Figure with the different dependent variables).

Instance 2, previously described in Section 2.1 was considered to assess the trained models. The evaluation of the instance follows $V > 63$, 411 L, $\text{DO} < 13.377 \text{ mg/L}$, $F_s > 82 \text{ L/h}$, $\text{DO} > 11.634 \text{ mg/L}$, and $\text{O}_{2, \text{og}} < 0.19\%$ which leads to LM 8:

$$P = -0.0042\text{DO} + 0.0023\text{CO}_{2, \text{og}} + 1.7966\text{O}_{2, \text{og}} + 16.9941 \\ = 17.28 \text{ g/L}$$

By substituting the respective values of DO , $\text{CO}_{2, \text{og}}$, and $\text{O}_{2, \text{og}}$, it is obtained a value of 17.28 g/L for P , which is slightly lower than the simulated value with the *IndPenSim* model, 19.629 g/L (Fig. 4B). The same procedure is used with the instances 1 and 3, by applying LM 11 and LM6 for which the values of penicillin concentration corresponded to 11.708 and 24.535 g/L, respectively.

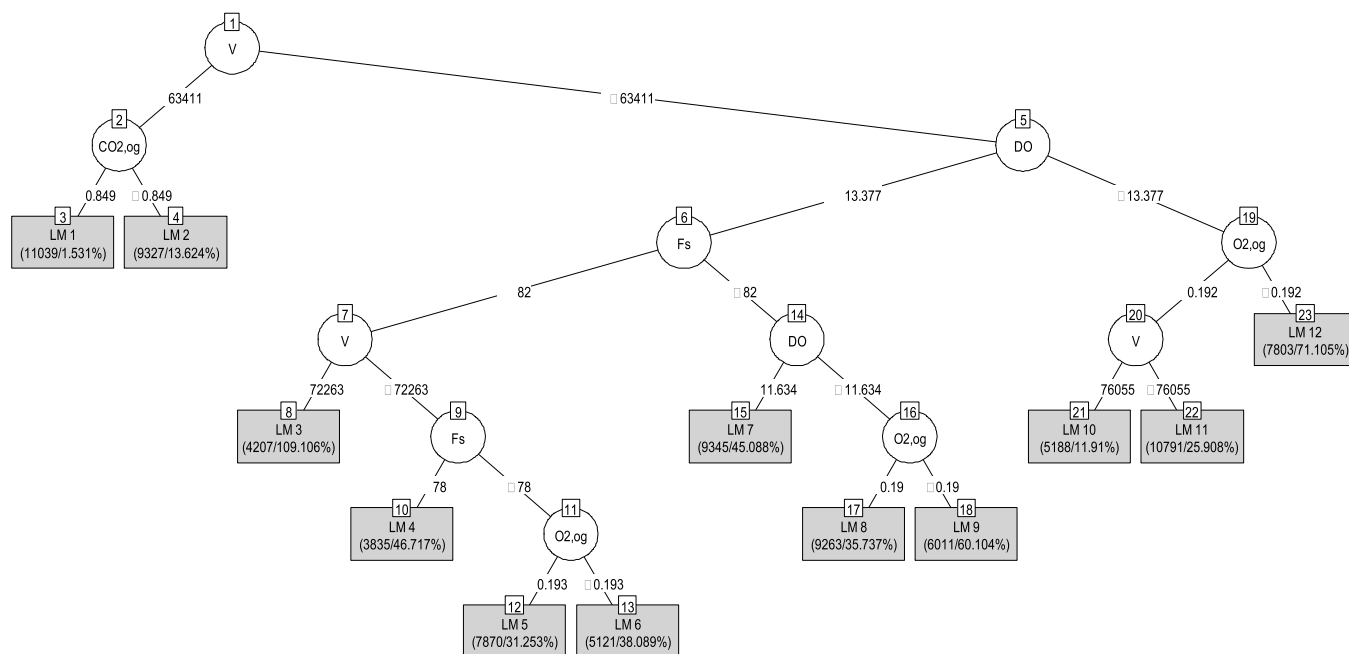


Fig. 5. M5 decision tree model.

LM 8 can be interpreted as follows: A value of $-0.0042 DO$ indicates a negative effect of DO over penicillin concentration, which means that an increase in DO could lead to a small decrease in P . The positive sign for $CO_{2,og}$ and $O_{2,og}$ indicate a positive effect on penicillin concentration, with the latter having a greater impact that does not go beyond 0.5 g/L , which can be obtained with the maximum $O_{2,og}$ reported in the cultures.

3.1.3. CUBIST

CUBIST has the same hyperparameter as M5 "minimum number of instances per leaf" because CUBIST is an extension of the M5 decision tree. In this case, a minimum number of instances per leaf of 12,000 has been selected as for M5. In addition, the hyperparameters committees and instance-based corrections were tuned. The number of committees refers to the number of individual CUBIST models generated. The use of committees in CUBIST is part of an ensemble method where iterative model trees are created in sequence, and subsequent trees are created using adjusted versions of the training set outcome (Kuhn and Johnson, 2013). On the other hand, instance-based corrections refer to nearest neighbors to adjust the predictions from the CUBIST model. Once CUBIST predicts an instance, instance-based corrections can find their nearest neighbors and determine the average of these training set points (Kuhn et al., 2012). A sequence of committees between 1 and 10 were tested and 0, 1, 3, 5 and 7 neighbors were analyzed for instance-based corrections as shown in Fig. A.7 in Appendix.

CUBIST was tuned with just one committee and three neighbors to prioritize model interpretability. CUBIST created a set of 20 rules and all independent variables were used to build the regression models. Table 3 presents the percentage distribution of the independent variables in CUBIST's rules and regression models. Table A.3 in Appendix lists all the rules and linear models generated by CUBIST.

Both conditional and regression models present an explicit use of variables such as F_s , V , $CO_{2,og}$, DO and $O_{2,og}$ with values between 36 and 88 % in the conditional rules and 91 and 100 % in the regression models. The variables F_s , V , and $O_{2,og}$ are 88, 80, and 71 % in the conditional models, which indicates that these variables are the most important in the construction of the cubist trees. On the other hand, the same variables represent 93, 100, and 91 % in the regression models, which means that these variables are representative for model predictions. Variables

Table 3

Percentage of the usage of attributes in the conditionals and linear regressions of the CUBIST model. The statistics of the linear models and rules were computed by CUBIST R Package. Additional statistics of the CUBIST model are provided in Table A.3.

Variable	Conditional (%)	Model (%)
F_s	88	93
V	80	100
$O_{2,og}$	71	91
DO	66	92
$CO_{2,og}$	36	94
pH	0	51
T	0	38

such as pH and T indicate 0 % of usage in the conditional models. Nevertheless, the same variables report values of usage of 51 and 38 % in the regression model, which correspond to the critical importance of environmental conditions in cell growth, penicillin formation, and consequently in penicillin hydrolysis (Goldrick et al., 2019). Agitation reports a constant value throughout the entire culture (100 rpm), the percentage of attribute usage in the conditionals and linear regressions is 100 %. As a result, it is omitted from the analysis.

For the sake of understanding, let us consider instance 3, previously described in Section 2.1, to assess the capability of the trained models. The evaluation of the instance follows Rule ID 9 (Table A.3 in Appendix):

If $DO > 11.295$ and $CO_{2,og} 1.4316$ and $O_{2,og} > 0.19108$ and $V > 68,944$ and $75 < F_s < 80$

The regression applied is as follows:

$$P = 375.7967 + 4.505F_s + 22.4CO_{2,og} - 930O_{2,og} - 2.67 DO + 1 \times 10^{-4}V - 1.8 T$$

This rule predicts values for penicillin concentration between 4.847 and 31.820 g/L, with an estimated error of 3.242 g/L. Since the volume varies between 56 and $90 \times 10^3 \text{ L}$ and in this case, it is around $69 \times 10^3 \text{ L}$, the expected value of penicillin would be small. The simulated value with the *IndPenSim* model is 29.104 g/L and the predicted with the CUBIST model is 26.251 g/L (Fig. 4C). It is worth noting that the value

from *IndPenSim* is within the estimated error of the regression model ID 9. This model indicates a larger contribution of $O_{2,og}$ and V which are compensated by DO and the independent coefficient. The same procedure was used with instances 1 and 2 by applying ID 5 and ID 9. Values of 8.684 and 15.795 g/L were obtained for instances 1 and 2, respectively.

Different from the M5 and CART, the rules set for CUBIST split the data without hierarchy, making the interpretation and learning from the simulations more difficult. For instance, rule ID 19 can predict P values from 9.080×10^{-5} to 36.183 g/L, so there cannot be a clear data separation.

3.1.4. Random forest

In a random forest, the hyperparameter to be estimated is Ntrees that refers to the number of decision trees included in the random forest. Decision trees were tested between 1 and 50, as shown in Table A.2 and Fig. A.8 in the Appendix. Increasing the number of trees in the random forest can improve model performance. However, having multiple decision trees make the interpretation of the model difficult compared to a single decision tree. RF was tuned with 5 trees in the forest, prioritizing the enhancement transparency/interpretability and the values obtained for MAE, RMSE and R^2 .

Fig. 6 depicts tree number 1 in the forest, where variables such as V , $O_{2,og}$, F_s , and DO were used. The rest of the forest are displayed in Fig. A.1 to A.4 in Appendix. The principle is similar to CART decision tree; however, the five trees are evaluated at the same time and their average value is reported as the predicted value. The five obtained trees are quite similar in the given rules, where only small variations of the decision variables could lead to a different prediction. The only difference is observed in tree 5 for instances whose characteristics follow $V > 68,924$, $DO < 13.33$ mg/L and $F_s < 82$ L/h where the final rule depends on $O_{2,og}$ instead of F_s . An instance with a value of $68,417 < V(78,434$ L, $DO)13.393$ mg/L and $O_{2,og} \leq 0.193\%$ could predict a mean value of P of 8.294 mg/L, which is near to the value 8.316 g/L obtained with the *IndPenSim* model (Fig. 4D). This is the case of instance 1 previously described in Section 2.1. Similar to CART, values with low $V < 63,000$ lead to low P titers which are the lowest if $O_{2,og} < 0.985\%$. On the contrary, large titers of P , for example 36.1 g/L can be obtained with $62,652 \leq V \leq 68,418$ L, $DO \leq 9.689$ mg/L and $F_s > 77.5$ L/h. In this case, volumes are low because of sampling along the culture.

3.2. Comparison of soft sensors based on interpretable learners

Four interpretable models have been constructed with the optimal parameters mentioned in the previous section. The training and test sets consist of 79 and 21 batches (Table 1), respectively, equally selected among the four control strategies. Table 4 shows the performance evaluation of the models with the test sets by control strategy and with the overall test set. Four statistical measures were used to evaluate the prediction performance of the models: Mean Absolute Error (MAE), Squared Mean Error (SME), Root Mean Squared Error (RMSE), and the

Table 4

Results of the interpretable learners evaluated with a test set divided by control strategies: (i) Controlled by recipe-driven approach, (ii) Controlled by operators, (iii) Controlled by an APC solution using the Raman spectroscopy and (iv) faults resulting in process deviations. Statistical criteria (SC) used to estimate the performance of regression learners: R^2 , MSE, RMSE, and MAE.

Learners	Test set	Statistical criteria			
		R^2	MSE	RMSE	MAE
CART decision tree	Controlled by recipe driven approach	0.838	17.576	4.192	2.998
	Controlled by operators	0.776	23.382	4.835	3.497
	Controlled by an APC solution using the Raman spectroscopy	0.913	9.642	3.105	2.266
	Faults resulting in process deviations	0.535	88.073	9.384	6.947
	Overall test sets	0.775	24.061	4.905	3.328
M5 decision tree	Controlled by recipe driven approach	0.837	17.789	4.217	2.807
	Controlled by operators	0.775	23.741	4.872	3.411
	Controlled by an APC solution using the Raman spectroscopy	0.897	11.309	3.362	2.114
	Faults resulting in process deviations	0.579	79.290	8.904	7.186
	Overall test sets	0.777	23.823	4.880	3.221
CUBIST	Controlled by recipe driven approach	0.940	6.549	2.559	1.708
	Controlled by operators	0.866	14.009	3.742	2.444
	Controlled by an APC solution using the Raman spectroscopy	0.979	2.231	1.493	1.001
	Faults resulting in process deviations	0.699	31.330	5.597	3.768
	Overall test sets	0.908	9.916	3.149	1.920
RF	Controlled by recipe driven approach	0.940	6.466	2.542	1.641
	Controlled by operators	0.883	11.891	3.448	2.239
	Controlled by an APC solution using the Raman spectroscopy	0.973	2.909	1.705	1.106
	Faults resulting in process deviations	0.716	38.406	6.197	4.334
	Overall test sets	0.905	10.202	3.194	1.928

coefficient of determination (R^2).

Table 4 reports the assessment of soft sensors applied for industrial-scale fed-batch fermentation simulations of penicillin production. As it can be observed, the test set controlled by recipe driven approach presented the best results using RF learners, with a high R^2 and low metric errors MSE, RMSE, MAE, where the values obtained were 0.940, 6.466, 2.542, and 1.641, respectively. The nature of the RF ensemble and the rule-based approach of CUBIST contribute to the ability to handle the complexity of bioprocesses under these conditions. On the other hand, CUBIST has demonstrated a high accuracy when utilizing the data set of the controlled by operator's technique, even if its performance was lower for the Controlled by recipe-driven approach. The obtained values

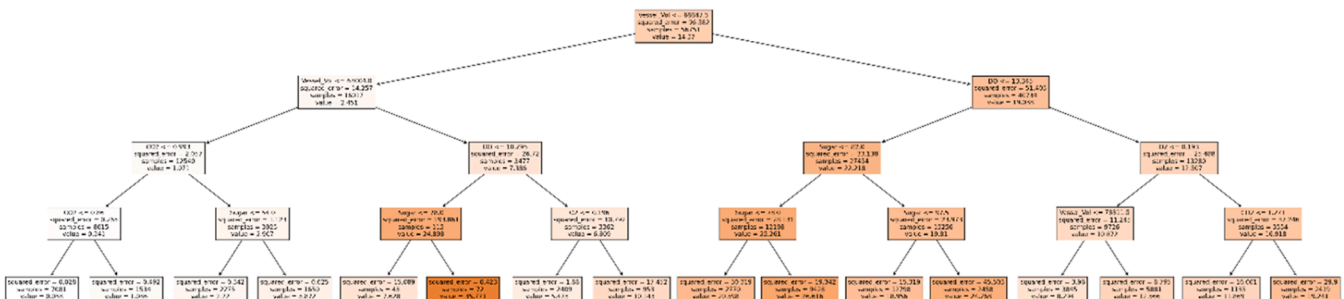


Fig. 6. Tree number 1 in the forest (Max tree depth = 7).

for R^2 , MSE , $RMSE$, and MAE were 0.883, 11.891, 3.448, and 2.239. This suggests that this model is sensitive to errors introduced by human operators in the execution of the bioprocess. The strategy Controlled by an APC solution using the Raman spectroscopy was well predicted with a high degree of accuracy when RF and CUBIST learners were used. Their R^2 , MSE , $RMSE$, and MAE values were in the range of 0.973–0.979, 2.231–2.909, 1.493–1.705, and 1.001–1.106, respectively. This implies that using cutting-edge sensor technologies for bioprocess monitoring and control is a good fit for these models. The strategy Faults resulting in process deviations was more challenging to predict with the four learners; however, CUBIST and RF displayed a good performance with values between 0.699–0.716, 31.330–38.406, 5.597–6.197, and 3.768–4.334 for R^2 , MSE , $RMSE$, and MAE .

Based on the overall test sets, the CUBIST learner achieved remarkable results for predicting P with all the control scenarios reporting values of 0.908, 9.916, 3.149 y 1.920 in R^2 , MSE , $RMSE$, and MAE . Results indicate that the model represents well a large part of the variability observed in penicillin concentration and is a good fit for model predictions. The rule-based nature of CUBIST seems beneficial for preserving the complex interactions in the bioprocess data. Similarly, this occurs with RF, where values of 0.905, 10.202, 3.194, and 1.928 were obtained for R^2 , MSE , $RMSE$, and MAE . As an ensemble method, RF also performed well throughout the different control scenarios, demonstrating its capacity for generalization and process condition adaptation.

On the other hand, CART and M5 were the models with the lowest prediction performance for all test sets. Predictions generated by CART lack smoothness and continuity due to a fixed penicillin production in the leaves, which did not change for each branch of the tree. For instance, CART approximates the penicillin concentration using piecewise constant segments, as seen in Fig. 7. M5 trees address the problem of discontinuity of CART by fitting piecewise linear models. However, in linear regressions, the most important values correspond to the intercepts with only a small variation due to the other dependent variables, which contributed to around 1 g/L difference from the intercept. Hence, M5 has a quite linear behavior, as displayed in Fig. 7.

Both learners present advantages in their application to penicillin production. CUBIST allows to determine which variables are the most

used for training the models and presents a wide set of rules that allows greater precision when calculating penicillin concentration. On the other hand, the RF presents a higher degree of explainability with some variables that are in the rules/conditions for separating the data. This allows to obtain very approximate values of penicillin concentrations. This is consistent with the dynamic *IndPenSim* model used to obtain the dataset, which was primarily focused on monitoring penicillin concentration rather than fault detections.

Fig. 7 illustrates the predictions of the soft sensors for monitoring penicillin production. Specifically, simulations of batches 15, 60, and 86 have been conducted based on different experiments: controlled by a recipe-driven approach, by operators, and by an APC solution using Raman spectroscopy, respectively. Predictions generally follow the dynamics of Penicillin production; however, the maximum values are not well predicted by any of the models, especially for the data where operators controlled the production.

The use of the algorithms for the data set considering process fault detection is displayed in Fig. 7 on Batch 91. All the models fall short when estimating values of this type of strategy. This could be related to the fact that training has been performed with data from the four strategies where only 8 batches were used for this strategy compared to the 71 batches used for the rest.

An important aspect to consider in interpretable models is the information they can provide about biological process. For example, let us consider batch 15 presented in Fig. 7, which was selected to demonstrate that these models can provide further insights of the process that can be complemented with expert knowledge. In this batch, it is observed that the models perform badly between 100h to 135h. Let us take the instance around 110h where it is observed that all models overestimate protein concentration with respect to the obtained with *IndPenSim*, 16.5190 g/L. Protein values of 19.92 g/L and 21.83 g/L were obtained by applying the CART decision tree and rule LM7 of the M5 decision tree. According to the trees presented in CART and M5, the variables with the highest importance are V , DO , and $O_{2,og}$. This mismatch is explained further in Fig. A.9 in the Appendix which presents the behavior over time of all the variables used for the training. The batches corresponding to the minimum and maximum values for all variables were displayed and corresponded to batches 4 and 30. In Fig. A.9, it can be observed that

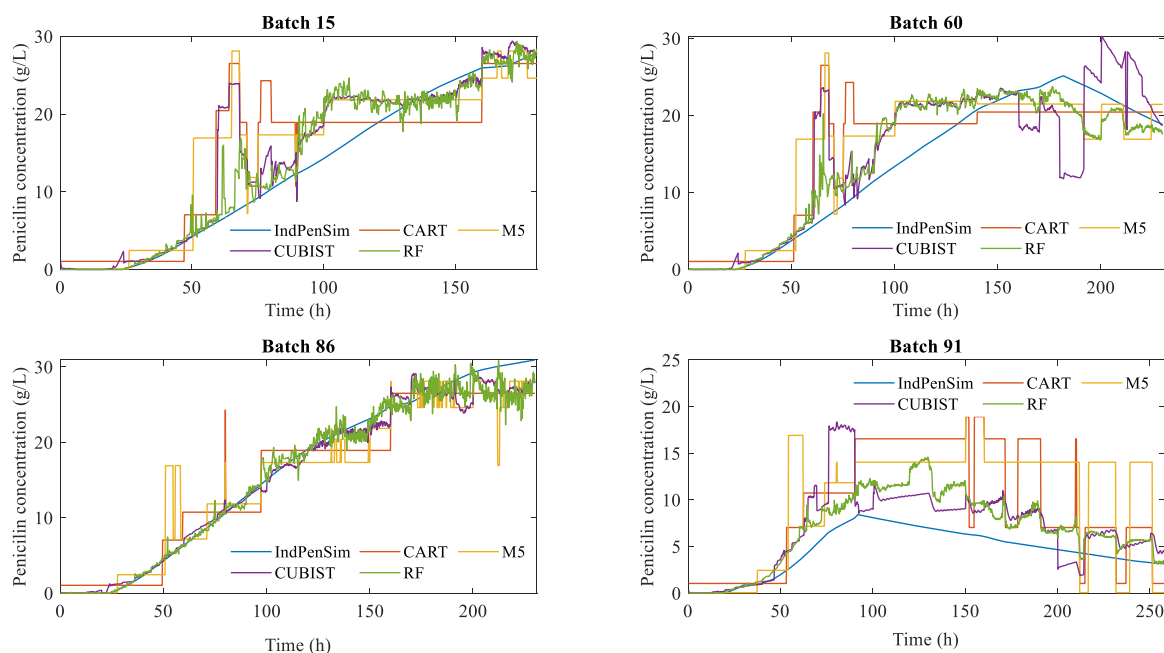


Fig. 7. Penicillin concentration of *IndPenSim* and prediction of the interpretable models for batches 15, 60, 86, and 91. Batches belong to a different control strategy. Control by recipe-driven approach: batch 15, control by operators: batch 60, control by an APC solution using Raman spectroscopy: batch 86, and faults resulting in process deviations: batch 91.

variables such as V , F_s , agitation, T , and pH have a similar behavior for batch 15. However, the values of $CO_{2,og}$, $O_{2,og}$, and DO are outside this min-max range, especially DO which is lower, and thus, the prediction of the models is not correct. Therefore, the overestimation can tell us where the error is coming from and eventually the model could be used to say that when DO is lower than a certain range, protein concentration will be overestimated. This error could be corrected if batch 15 is considered in the training.

4. Conclusions and future works

In this paper, it is proposed and comparative assessed a set of interpretable models to develop soft sensors for the industrial-scale fed-batch fermentation of penicillin production. Among the four proposed models, CUBIST and RF learners provide the best results in terms of R^2 , $RSME$, and MAE . These results are based on the rule-based approach of CUBIST that contributes to its ability to handle the complexity of bioprocesses and the ensemble nature of the RF to deal with small variations by constructing different decision trees.

The ability to understand and explain the relationships between independent variables and penicillin concentration facilitates the identification of actionable insights for bioprocess improvement. Nevertheless, the interpretation of the models must be accompanied by expertise in the area. Furthermore, CUBIST and RF models could be used to explore the variations within the batches for detecting human errors or faults associated with the penicillin production, as it has been evidenced in batches 60 and 91 in Fig 7.

In terms of transparency degree, the four learners can vary depending on hyperparameters values defined for each learner. For example, CART has a high transparency degree considering a not-so-high value for maximum number of decision tree levels, if the tree has many levels, it will be more difficult to understand. As observed in Fig. 3, the CART model reached a maximum depth of 5 levels, which allows to understanding how the model works.

On the other hand, M5, CUBIST, and RF are more sophisticated transparent learners that have more hyperparameters to tune, such as the number of rules, linear regressions, decision trees, and committees. Increasing the values of these hyperparameters leads to lower transparency in the models, yet they exhibit higher precision as is observed in Table 4. For instance, the M5 model contains 12 model regressions (Table A.1.), CUBIST model consists of 20 rules and 20 linear regressions (Table A.3.) and RF comprises a forest of 5 CART trees with a maximum depth of 5 levels (Figs. 3, A.1–A.4). Although these learners have a lower level of transparency compared to CART, they are interpretable, and each of their rules, linear regressions, and trees can be analyzed and understood, providing insight into how these models work.

As future work, the integration of the nature of the data to train the different interpretable learners could be explored, e.g., how to improve

the CUBIST and the RF prediction using the control strategies: control by operators and the detection of faults resulting in process deviations.

In addition, we propose as future works the exploration of integrating post-hoc explanations techniques with (i) data-driven black-box models and (ii) hybrid models that combine first-principles physics-based dynamics with data-driven models. Black-box models based on deep learning architectures have gained attention in process monitoring, control, and optimization (Sitapure and Kwon, 2023b), while hybrid models are considered a 'best-of-both-worlds' approach due to their direct integration of system-agnostic first principles with system-specific data-driven parameters (Sitapure and Kwon, 2023a). This latest approach has been used in several studies combining first principles with deep neural networks in different fermentation processes (Bang et al., 2022; Pinto et al., 2022; Shah et al., 2021).

CRediT authorship contribution statement

Juan Camilo Acosta-Pavas: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **Carlos Eduardo Robles-Rodriguez:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **David Griol:** Writing – review & editing, Validation, Supervision, Conceptualization. **Fayza Daboussi:** Writing – original draft, Validation, Supervision. **Cesar Arturo Aceves-Lara:** Writing – review & editing, Visualization, Supervision, Formal analysis, Conceptualization. **David Camilo Corrales:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: David Camilo Corrales reports a relationship with French National Institute for Agricultural Research INRAE that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was funded (or co-funded) by the European Union under the Horizon Europe project Bioindustry 4.0, grant n. 101094287.

Appendix

Table A.1
Linear regressions of the M5 model.

LM	Linear Model (LM)
1	$P = -0.002 DO + 0.0123 CO_{2,og} + 0.8542 O_{2,og} + 0 V + 0 F_s - 0.121$
2	$P = -0.002 DO + 0.0131 CO_{2,og} + 0.8542 O_{2,og} + 0 V + 0 F_s + 2.2669$
3	$P = -0.0031 DO + 0.0023 CO_{2,og} - 0.6405 O_{2,og} + 0 V + 0.0011 F_s + 16.8602$
4	$P = -0.0031 DO + 0.0023 CO_{2,og} - 2.4304 O_{2,og} + 0 V + 0.001 F_s + 21.8325$
5	$P = -0.0031 DO + 0.0023 CO_{2,og} - 2.4154 O_{2,og} + 0 V + 0.0005 F_s + 28.4855$
6	$P = -0.0031 DO + 0.0023 CO_{2,og} - 3.5271 O_{2,og} + 0 V + 0.0005 F_s + 25.2126$
7	$P = -0.0051 DO + 0.0023 CO_{2,og} + 1.4571 O_{2,og} + 0 V + 0 F_s + 21.5628$
8	$P = -0.0042 DO + 0.0023 CO_{2,og} + 1.7966 O_{2,og} + 0 V + 0 F_s + 16.9941$

(continued on next page)

Table A.1 (continued)

LM	Linear Model (LM)
9	$P = -0.0042 DO + 0.0023 CO_{2,og} + 2.1032 O_{2,og} + 0 V + 0 F_s + 19.9982$
10	$P = -0.0026 DO + 0.0023 CO_{2,og} + 1.1926 O_{2,og} + 0 V + 0 F_s + 6.8134$
11	$P = -0.0026 DO + 0.0023 CO_{2,og} + 1.1926 O_{2,og} + 0 V + 0 F_s + 11.5183$
12	$P = -0.0026 DO + 0.0023 CO_{2,og} + 1.755 O_{2,og} + 0 V + 0 F_s + 13.612$

Table A.2

Tunning the trees of Random Forest (Number of resamples: 30).

Metric	Ntree	Min	1st Qu	Median	Mean	3rd Qu	Max
R^2	1	0.957	0.970	0.973	0.972	0.975	0.979
	5	0.986	0.988	0.989	0.989	0.990	0.991
	10	0.990	0.990	0.991	0.991	0.991	0.993
	15	0.991	0.991	0.992	0.992	0.992	0.993
	20	0.991	0.992	0.992	0.992	0.993	0.993
	25	0.991	0.992	0.992	0.992	0.993	0.993
	30	0.991	0.992	0.993	0.993	0.993	0.993
	35	0.991	0.992	0.993	0.993	0.993	0.994
	40	0.992	0.992	0.993	0.993	0.993	0.994
	45	0.992	0.993	0.993	0.993	0.993	0.994
	50	0.992	0.993	0.993	0.993	0.993	0.994
	1	0.507	0.558	0.598	0.619	0.669	0.827
	5	0.436	0.464	0.486	0.487	0.503	0.571
	10	0.412	0.441	0.453	0.453	0.469	0.499
	15	0.401	0.431	0.442	0.439	0.450	0.467
MAE	20	0.403	0.423	0.428	0.430	0.439	0.461
	25	0.398	0.416	0.424	0.424	0.433	0.454
	30	0.391	0.412	0.417	0.419	0.427	0.446
	35	0.392	0.410	0.418	0.417	0.424	0.442
	40	0.387	0.408	0.416	0.415	0.424	0.437
	45	0.391	0.406	0.413	0.413	0.422	0.433
	50	0.389	0.403	0.414	0.411	0.419	0.430
	1	1.417	1.526	1.603	1.631	1.68	2.027
	5	0.908	0.966	1.00	1.00	1.034	1.158
	10	0.813	0.880	0.903	0.899	0.933	0.975
	15	0.798	0.838	0.864	0.862	0.886	0.932
	20	0.775	0.817	0.842	0.842	0.862	0.899
	25	0.772	0.808	0.829	0.828	0.841	0.886
	30	0.769	0.797	0.819	0.819	0.836	0.881
	35	0.760	0.794	0.813	0.815	0.832	0.890
$RMSE$	40	0.760	0.789	0.809	0.810	0.828	0.875
	45	0.754	0.786	0.807	0.806	0.825	0.865
	50	0.748	0.784	0.805	0.803	0.821	0.858

Table A.3

Rules and linear regressions of the CUBIST model. Committees = 0 and rules = 20.

Rule ID	Rule	Regression model	Cases	Mean	Range	Est error
1	IF $V \leq 59,188$ AND $F_s \leq 76$	$P = -2.5540279 + 0.9 CO_{2,og} + 4e-005 V$	7803	0.011	3.6947e-026 to 1.2413	0.093
2	IF $CO_{2,og} \leq 1.004$ AND $F_s \leq 76$	$P = -48.2768694 + 0.00033 V + 2.8 CO_{2,og} + 125 O_{2,og} + 0.022 F_s + 0.13 DO$	12,421	0.306	3.6947e-026 to 6.5143	0.112
3	IF $CO_{2,og} > 1.004$ AND $O_{2,og} \leq 0.19462$ AND $V \leq 68,944$ AND $F_s \leq 76$	$P = -17.1947906 + 0.098 F_s + 4.6 CO_{2,og} - 5e-005 Vessel Vol. + 61 O_{2,og} + 0.1 DO$	9895	3.931	0.00093733 to 7.8476	0.305
4	IF $CO_{2,og} \leq 1.3519$ AND $V \leq 68,944$ AND $F_s > 76$	$P = -81.5016775 + 8 CO_{2,og} + 0.00021 V - 0.069 F_s - 0.65 DO + 9 pH + 120 O_{2,og}$	2801	4.234	0.00089398 to 18.981	1.022
5	IF $DO > 13.402$ AND $O_{2,og} \leq 0.19108$ AND $V > 68,944$	$P = -96.9497476 + 0.00041 V - 1.24 DO + 382 O_{2,og} + 3.4 CO_{2,og} + 0.028 F_s + 2 pH$	12,846	10.694	4.3084 to 28.26	1.115
6	IF $DO > 14.515$ AND $V > 68,944$ AND $F_s \leq 65$	$P = -230.0167296 + 11.4 CO_{2,og} + 1205 O_{2,og} + 0.125 F_s - 0.58 DO - 6e-005 V$	238	11.649	7.2002 to 21.231	0.242
7	IF $CO_{2,og} > 1.004$ AND $O_{2,og} > 0.19462$ AND $V \leq 68,944$ AND $F_s \leq 76$	$P = -444.7833835 + 1668 O_{2,og} + 0.00086 V - 0.116 F_s - 1 DO + 14 pH + 0.2 CO_{2,og}$	265	12.342	2.6954 to 18.563	1.000
8	IF $DO > 13.573$ AND $O_{2,og} > 0.19108$ AND $V > 68,944$ AND $F_s > 75$ AND $F_s \leq 80$	$P = -587.9608979 + 0.357 F_s + 16.4 CO_{2,og} - 4.5 DO + 0.00066 V + 910 O_{2,og} + 1.3 T$	2279	12.457	4.8473 to 30.783	2.552
9	IF $DO > 11.295$ AND $CO_{2,og} \leq 1.4316$ AND $O_{2,og} > 0.19108$ AND $V > 68,944$ AND $F_s > 75$ AND $F_s \leq 80$	$P = 375.796699 + 4.505 F_s + 22.4 CO_{2,og} - 930 O_{2,og} - 2.67 DO + 0.0001 V - 1.8 T$	3275	16.932	4.8473 to 31.82	3.242
10	IF $DO > 11.295$ AND $DO \leq 13.402$ AND $O_{2,og} \leq 0.19108$ AND $V > 68,944$ AND $V \leq 75,553$ AND $F_s \leq 80$	$P = -115.9511318 + 0.675 F_s - 9.52 DO + 0.00144 V - 25.5 CO_{2,og} + 20 pH + 29 O_{2,og}$	383	18.218	5.8616 to 34.233	3.067

(continued on next page)

Table A.3 (continued)

Rule ID	Rule	Regression model	Cases	Mean	Range	Est error
11	IF $DO > 11.295$ AND $O_{2,og} > 0.19108$ AND $V > 68,944$ AND $F_s > 80$	$P = -71.2189295 + 18.7 CO_{2,og} + 0.235 F_s - 2.26 DO + 342 O_{2,og} + 2e-005 V$	7457	18.676	6.1592 to 34.987	3.182
12	IF $DO \leq 13.402$ AND $O_{2,og} \leq 0.19108$ AND $V > 68,944$ AND $F_s > 80$	$P = 1985.4042782 + 1430 O_{2,og} + 11.6 CO_{2,og} + 0.214 F_s - 1.67 DO - 7.5 T - 3e-005 V - 3 pH$	19,061	19.176	7.5064 to 35.088	2.068
13	IF $DO \leq 11.295$ AND $O_{2,og} > 0.19253$ AND $F_s > 80$	$P = 572.3606495 - 2629 O_{2,og} + 0.346 F_s + 14.3 CO_{2,og} - 0.00059 V - 8 pH$	409	19.909	12.672 to 30.377	1.226
14	IF $O_{2,og} > 0.19108$ AND $V > 68,944$ AND $F_s > 65$ AND $F_s \leq 75$	$P = -541.432649 + 10.023 F_s - 2602 O_{2,og} - 0.00078 V + 2.52 DO - 2.5 CO_{2,og} + 1.1 T + 3 pH$	2514	19.968	4.1505 to 32.518	2.632
15	IF $DO \leq 14.515$ AND $V > 68,944$ AND $F_s \leq 65$	$P = 169.4260538 + 27.6 CO_{2,og} - 0.00083 V + 2.94 DO - 854 O_{2,og} + 0.033 F_s$	2105	20.950	5.0614 to 30.629	2.620
16	IF $DO \leq 11.295$ AND $O_{2,og} \leq 0.19253$ AND $V > 68,944$ AND $F_s > 80$	$P = -154.3022432 + 0.273 F_s + 914 O_{2,og} + 6 CO_{2,og} - 0.00023 V - 2 pH - 0.08 DO$	6716	22.151	7.5064 to 34.681	2.106
17	IF $DO > 11.295$ AND $DO \leq 13.402$ AND $O_{2,og} \leq 0.19108$ AND $V > 75,553$ AND $F_s \leq 80$	$P = -69.2333804 + 1.396 F_s - 0.00035 V + 1.13 DO$	1455	24.735	15.208 to 32.464	1.460
18	IF $DO > 11.295$ AND $DO \leq 13.573$ AND $CO_{2,og} > 1.4316$ AND $O_{2,og} > 0.19108$ AND $F_s > 75$ AND $F_s \leq 80$	$P = 89.8862514 + 5.944 F_s - 1263 O_{2,og} - 6.2 CO_{2,og} - 0.00022 V - 5 pH + 0.24 DO - 0.8 T$	8611	26.167	6.3573 to 34.241	2.055
19	IF $CO_{2,og} > 1.3519$ AND $V \leq 68,944$ AND $F_s > 76$	$P = 322.0909392 - 1.071 F_s - 14.9 CO_{2,og} - 857 O_{2,og} - 0.94 DO - 0.00013 V$	214	26.827	0.00090802 to 36.183	1.103
20	IF $DO \leq 11.295$ AND $V > 68,944$ AND $F_s \leq 80$	$P = 531.4017384 + 1.418 F_s - 780 O_{2,og} - 0.00029 V + 0.28 DO - 1.5 T$	4454	28.098	6.2591 to 35.941	2.432

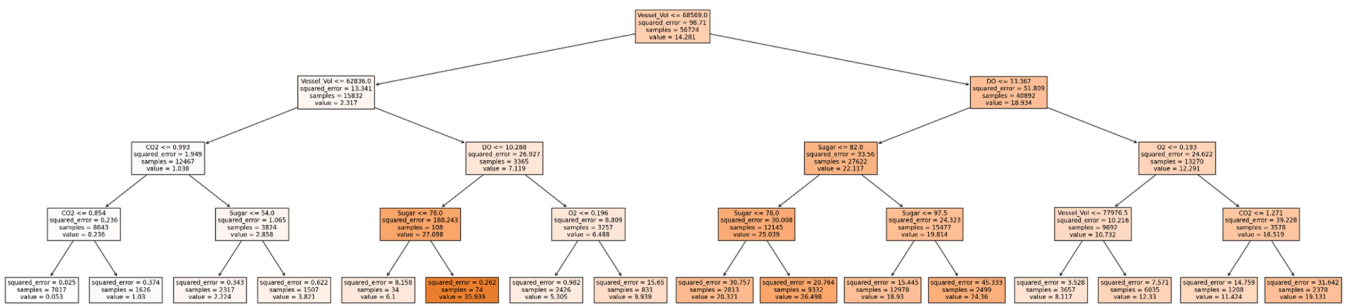


Fig. A.1. Tree number 2 in the forest. Max tree depth = 7.

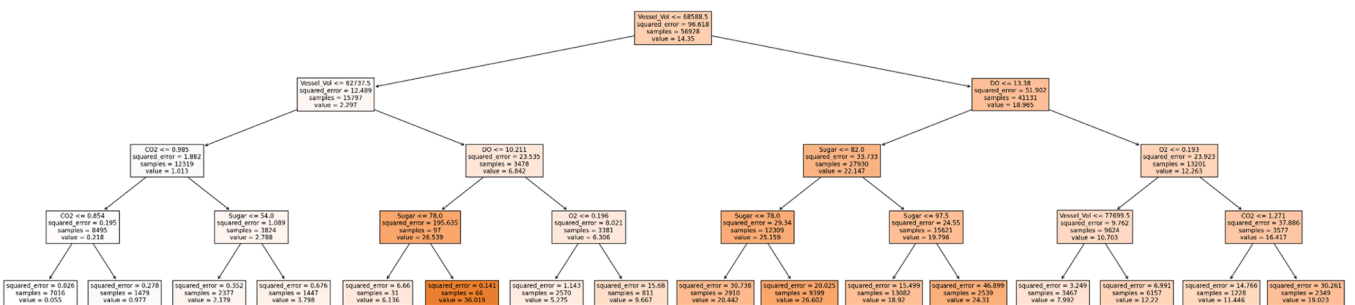


Fig. A.2. Tree number 3 in the forest. Max tree depth = 7.

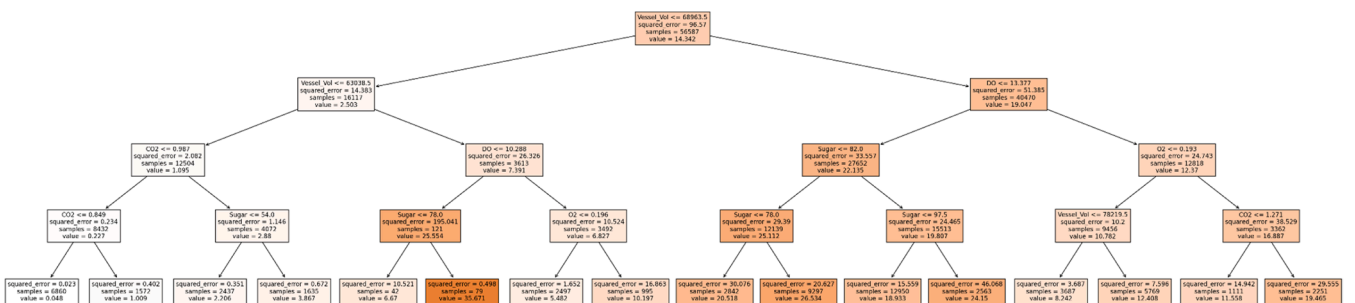


Fig. A.3. Tree number 4 in the forest. Max tree depth = 7.

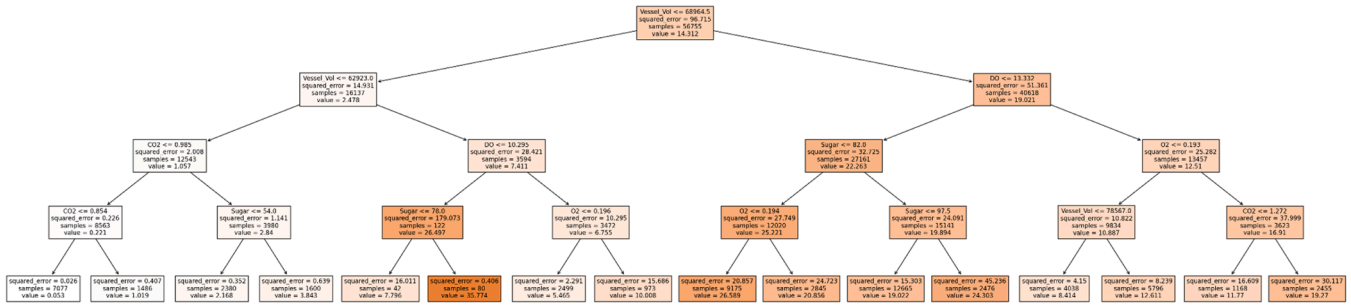


Fig. A.4. Tree number 5 in the forest. Max tree depth = 7.

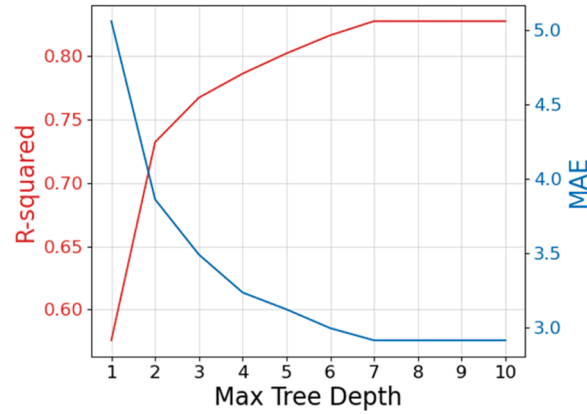


Fig. A5. Optimization of hyperparameter maximum tree depth in CART decision tree model.

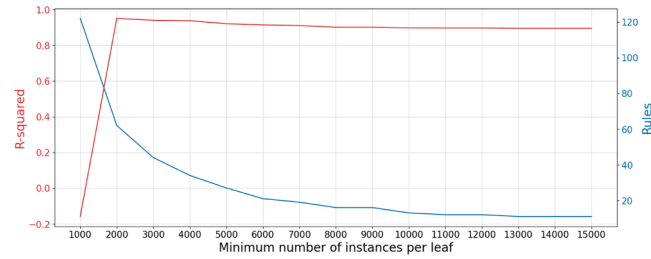


Fig. A.6. Optimization of hyperparameter - minimum number of instances per leaf = 12,000 inM5 decision tree model.

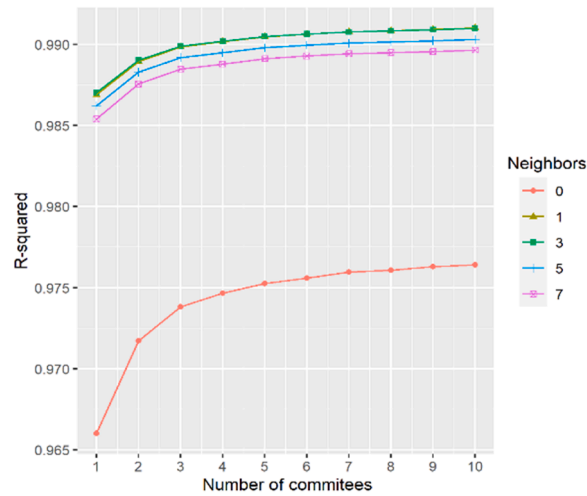


Fig. A.7. Optimization of CUBIST hyperparameters: committees and neighbors.

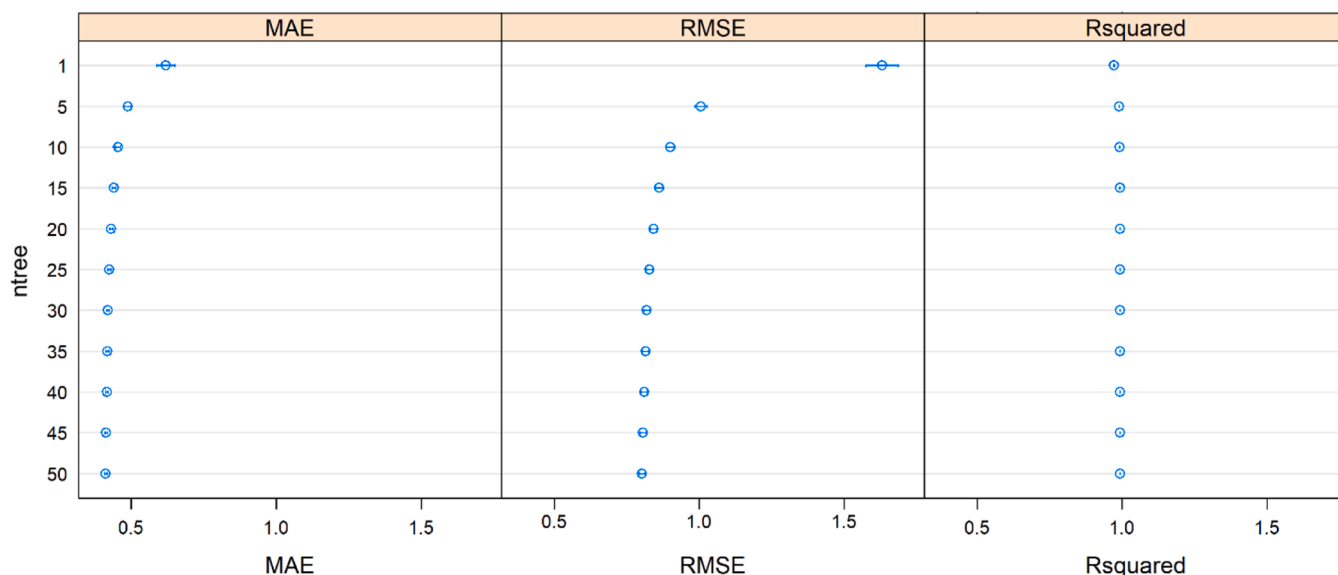


Fig. A.8. Tuning the trees of Random Forest. Ntrees = 5. Confidence level: 0.95.

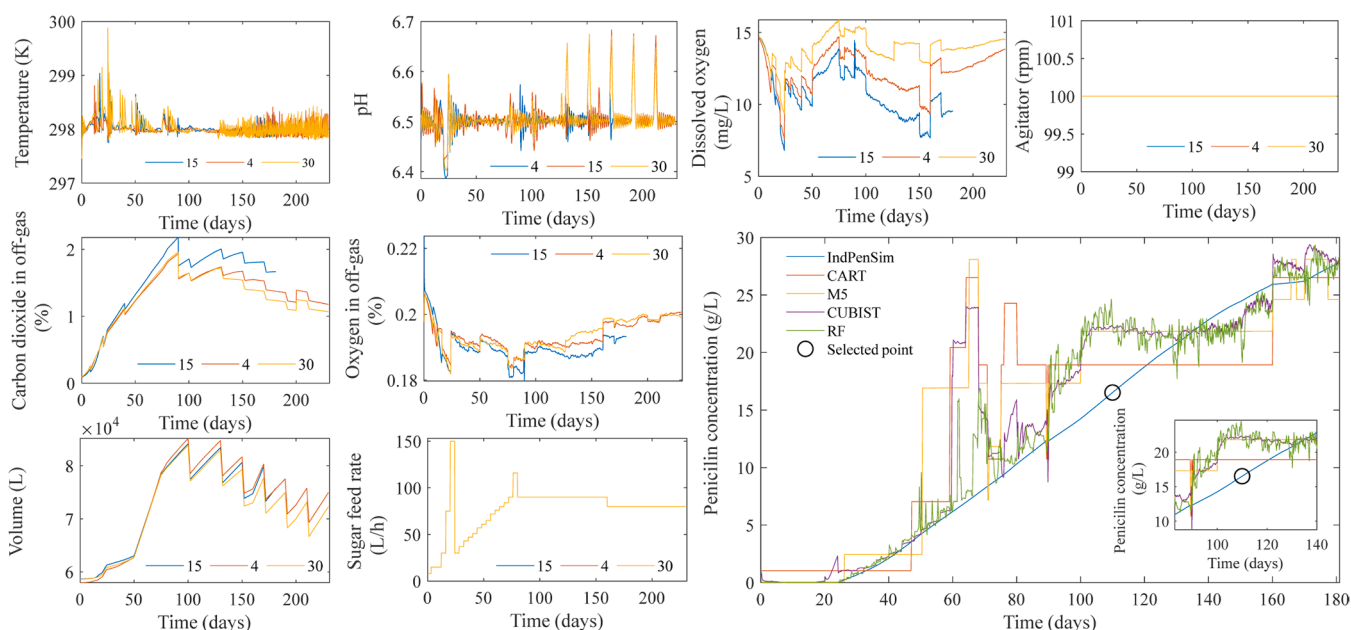


Fig. A.9. Dynamic behavior of independent and dependent variables in Batch 15.

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